

07 API REQUIREMENTS

API Requirements

A backend handoff sheet describing the minimum contracts needed by TRINETRA's MVP.

Endpoint requirement sheet

Method	Endpoint	Purpose	Access	Key input	Output
POST	/api/auth/register	Register a user	Public	Identity, contact, password	User ID / validation errors
POST	/api/auth/login	Authenticate a user	Public	Credentials	Session/token and role
GET	/api/disasters	List disasters	Authenticated	Filters, bounds, status	Disaster summaries
GET	/api/disasters/:id	Get disaster details	Authenticated	Disaster ID	Event, affected area, status
POST	/api/disasters	Create a disaster	Authority	Type, location, severity, details	Created disaster
GET	/api/alerts	List current alerts	Authenticated	Area/status filters	Alerts
POST	/api/alerts	Create/publish alert	Authority	Area, severity, instructions, validity	Published alert
GET	/api/resources	Get emergency resources	Authenticated	Location, radius, type	Resources with source/status
POST	/api/reports	Submit citizen report	Citizen	Type, location, description, image	Report ID/status
GET	/api/reports	View citizen reports	Authority	Status/location filters	Report list
GET	/api/risk/:location	Get risk information	Authority	Location + available features	Risk level/score/explanation

API conventions

- Use a versioned base path and consistent JSON naming.
- Validate all required fields, ranges, coordinates, and identifiers.
- Use predictable error envelopes and HTTP status codes.
- Apply role-based authorization; authentication alone is not sufficient for authority actions.
- Include IDs, timestamps, source/provenance, status, and pagination where relevant.
- Document rate limits, upload limits, and demo fallback fixtures.